

GUMBEL FLUCTUATIONS FOR THE COVER TIME OF NEAR-COLLISIONS IN DYSON BROWNIAN MOTION

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ABSTRACT. Let $\lambda(t)$ be a stationary Dyson Brownian motion, circular or of Ornstein–Uhlenbeck type, at inverse temperature $\beta > 1$, let τ_i be the first time the i -th spacing falls below ε , and let $\tau_{\text{cov}} = \max_i \tau_i$ be the cover time. The hazard of the i -th spacing is $\Lambda_i = \frac{2(\beta-1)}{\beta n} a_i \varepsilon^{\beta-1}$, where a_i is its collision coefficient, $\mathbb{P}(g_i \leq x) \sim a_i x^{\beta+1}/(\beta+1)$; for the circular ensemble a_i is a ratio of Morris integrals. We prove that when $\Lambda_i(\log n)^{2\beta+4} \rightarrow 0$ the normalized hitting times converge to a Poisson process with intensity $e^{-x} dx$, so that τ_{cov} has Gumbel fluctuations and each τ_i is asymptotically exponential. For the circular unitary ensemble this is unconditional: $\mathbb{P}(\frac{(n^2-1)\varepsilon}{24\pi} \tau_{\text{cov}} - \log n \leq x) \rightarrow e^{-e^{-x}}$ whenever $\varepsilon n^2 (\log n)^8 \rightarrow 0$. For other β and for the Hermite ensemble we assume a first-moment bound on the inverse square distances seen from a collision. Where the density varies over the index window, the centering acquires a $\log \log n$ correction. In the remaining regime, where $\varepsilon n^2 \rightarrow \infty$ but $n\varepsilon \rightarrow 0$, we show that the Gumbel law follows from a single static inequality for principal Dirichlet eigenvalues. The method is spectral: survival of a set of spacings is controlled by the principal Dirichlet eigenvalue of the union of their near-collision sets, computed to relative precision $o(1/\log n)$ from capacity bounds along orthogonal directions of configuration space.

1. INTRODUCTION

1.1. The question. In the normalization

$$(1) \quad d\lambda_k = \sqrt{\frac{2}{\beta n}} dB_k + \frac{1}{n} \sum_{j \neq k} \frac{dt}{\lambda_k - \lambda_j} - \frac{\lambda_k}{2} dt, \quad k = 1, \dots, n,$$

Dyson Brownian motion is stationary under the Hermite β -ensemble, whose empirical measure approaches the semicircle law $\varrho(x) = \frac{1}{2\pi} \sqrt{4 - x^2}$. The spacing $g_i = \lambda_{i+1} - \lambda_i$ is of order $1/(n\varrho)$, diffuses with coefficient $4/(\beta n)$, and is pushed away from zero by the drift $2/(ng_i)$: near zero, ng_i behaves on the time scale $1/n$ like a Bessel process of dimension $\beta + 1$, which for $\beta > 1$ never hits the origin.

Fix $\varepsilon > 0$, an index set I , and let

$$\tau_i = \inf\{t \geq 0 : g_i(t) \leq \varepsilon\}, \quad \tau_{\text{cov}} = \max_{i \in I} \tau_i.$$

The cover time τ_{cov} is the first time at which every spacing in I has come within ε of a collision. If near-collisions are rare and weakly dependent, the spacings behave like the coupons of a coupon collector, and τ_{cov} should have Gumbel fluctuations around $\log |I|$ divided by the rate. We prove this, identify the rate, and describe the extremal process of the late hitting times.

1.2. The hazard. Let p_i be the equilibrium density of g_i . The *collision coefficient*

$$(2) \quad a_i := \lim_{x \downarrow 0} x^{-\beta} p_i(x)$$

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exists and is finite and positive (Lemma 2.3). The *hazard* of the i -th spacing is

$$(3) \quad \Lambda_i := \frac{2(\beta-1)}{\beta n} a_i \varepsilon^{\beta-1},$$

the flux through the level ε of a Bessel process of dimension $\beta+1$, diffusion coefficient $4/(\beta n)$ and stationary density $a_i x^\beta$ near the origin. In the bulk of the Hermite ensemble $a_i \approx (\beta+1)\kappa_\beta(n\varrho(\gamma_i))^{\beta+1}$, with γ_i the classical location and κ_β the small-spacing constant of the Sine $_\beta$ process ($\kappa_2 = \pi^2/9$), so $\Lambda_i \asymp n^\beta \varepsilon^{\beta-1}$.

1.3. Main results. The circular Dyson process is

$$(4) \quad d\theta_k = \sqrt{\frac{2}{\beta n}} dB_k + \frac{1}{2n} \sum_{j \neq k} \cot \frac{\theta_k - \theta_j}{2} dt, \quad k = 1, \dots, n,$$

stationary under the circular β -ensemble. Its n cyclic spacings ($g_n = \theta_1 + 2\pi - \theta_n$) share one collision coefficient, a ratio of Morris integrals (Proposition 6.1):

$$(5) \quad a_\beta(n) = \frac{(n-1)\Gamma(1+\frac{\beta}{2})^n}{2\pi\Gamma(1+\frac{n\beta}{2})} \prod_{j=0}^{n-3} \frac{\Gamma(\frac{\beta}{2}j+2\beta+1)\Gamma(\frac{\beta}{2}(j+1)+1)}{\Gamma(\frac{\beta}{2}j+\beta+1)^2\Gamma(1+\frac{\beta}{2})}, \quad a_2(n) = \frac{n(n^2-1)}{24\pi}.$$

Theorem A (Circular unitary ensemble). *Start (4) with $\beta = 2$ from the circular unitary ensemble, let $\varepsilon_n n^2(\log n)^8 \rightarrow 0$, and put $\lambda_n = (n^2-1)\varepsilon_n/(24\pi)$. Then, with $\tau_{\text{cov}} = \max_{1 \leq i \leq n} \tau_i$,*

- (i) $\mathbb{P}(\lambda_n \tau_{\text{cov}} - \log n \leq x) \rightarrow \exp(-e^{-x})$ for every $x \in \mathbb{R}$;
- (ii) $\sum_{i=1}^n \delta_{\lambda_n \tau_i - \log n}$ converges in distribution, on $(x_0, \infty]$ for every $x_0 \in \mathbb{R}$, to the Poisson process with intensity $e^{-x} dx$;
- (iii) $\sup_{t \geq 0} |\mathbb{P}(\lambda_n \tau_i > t) - e^{-t}| \rightarrow 0$.

For general β and for the Hermite ensemble we need one estimate that we prove only for the circular unitary ensemble: a first-moment bound on the inverse square distances from a collision point to the other particles, under the law of the configuration conditioned on a collision (Hypothesis 2.4).

Theorem B (Circular ensemble). *Let $\beta > 1$, start (4) from the circular β -ensemble, assume Hypothesis 2.4 with $\kappa = 0$, and let $\lambda_n = \frac{2(\beta-1)}{\beta n} a_\beta(n) \varepsilon_n^{\beta-1}$ satisfy $\lambda_n(\log n)^{2\beta+4} \rightarrow 0$. Then (i)–(iii) of Theorem A hold.*

By Remark 6.2, $\lambda_n \asymp n^\beta \varepsilon_n^{\beta-1}$, so the condition is $\varepsilon_n \ll n^{-\beta/(\beta-1)}$ up to a power of $\log n$.

When the hazards differ, the centering is implicit. For an index set I_n let $t_n(x)$ solve

$$(6) \quad \sum_{i \in I_n} e^{-\Lambda_i t} = e^{-x}, \quad x > -\log |I_n|,$$

and let $x_n = t_n^{-1}$, with $x_n(0) = -\log |I_n|$.

Theorem C (Hermite ensemble). *Let $\beta > 1$, start (1) in equilibrium, let $I_n \subset \{1, \dots, n-1\}$, $|I_n| \rightarrow \infty$, satisfy Hypothesis 2.4, and let $\Lambda_{\max} = \max_{I_n} \Lambda_i$. Assume either $\kappa = 0$ and $\Lambda_{\max}(\log n)^{2\beta+4} \rightarrow 0$, or $\Lambda_{\max} \leq n^{-\delta}$ with $\delta \geq 4(\beta+1)\kappa$. Then $\sum_{i \in I_n} \delta_{x_n(\tau_i)}$ converges in distribution on $(x_0, \infty]$ to the Poisson process with intensity $e^{-x} dx$; in particular $\mathbb{P}(\tau_{\text{cov}} \leq t_n(x)) \rightarrow \exp(-e^{-x})$, and $\sup_t |\mathbb{P}(\tau_i > t) - e^{-\Lambda_i t}| \rightarrow 0$ uniformly in $i \in I_n$.*

For the bulk window $I_n = [n/4, 3n/4]$ of the Gaussian unitary ensemble the explicit centering is not $\log n$ (Corollary 7.2):

$$\mathbb{P}\left(\Lambda_{*} \tau_{\text{cov}} - \log n + \log \log n - \log c_* \leq x\right) \rightarrow \exp(-e^{-x}),$$

where $\Lambda_* = \frac{\pi^2}{3} \varrho(\gamma_*)^3 n^2 \varepsilon$ is the smallest hazard, attained at the two ends of the window, $\gamma_* \approx -0.8079$ is the first quartile of the semicircle law, and $c_* = (4 - \gamma_*^2)^{3/2} / (3\pi|\gamma_*|) \approx 0.8042$. The last spacings to be hit sit near the ends of the window, where the density is smallest; their number is of order $n/\log n$, not n .

In the regime $\varepsilon n^2 \rightarrow \infty$, $n\varepsilon \rightarrow 0$, near-collisions are still rare but more frequent than the relaxation of the spectrum. There the Gumbel law reduces to a static inequality.

Theorem D (Reduction). *Let $\beta > 1$, let I_n satisfy (P1)–(P2) of Hypothesis 2.4 with $\kappa = 0$, and let $n\varepsilon_n(\log n)^{\beta/(\beta-1)} \rightarrow 0$. If the principal Dirichlet eigenvalues of the sets $\bigcup_{i \in S} \{g_i \leq \varepsilon\}$, $S \subset I_n$, $|S| \leq m$, satisfy the lower bound (9) with $\eta_n \log n \rightarrow 0$ for every m , then $\mathbb{P}(\tau_{\text{cov}} \leq t_n(x)) \rightarrow \exp(-e^{-x})$, the same holds for every order statistic of the τ_i , and $\sup_t |\mathbb{P}(\tau_i > t) - e^{-\Lambda_i t}| \rightarrow 0$.*

The lower bound is a property of the equilibrium measure and the Dirichlet form only; no time scale enters. Lemma 8.1 reduces it further to an averaging inequality for an explicit conditional hazard (Conjecture 8.2).

1.4. Scope. The condition $\Lambda_i \ll (\log n)^{-2\beta-4}$ says that near-collisions are rarer than the relaxation of the whole spectrum, whose rate, the spectral gap, is a constant. The regime lies far below the static extremes: at a fixed time the smallest bulk spacing is of order $n^{-(\beta+2)/(\beta+1)}$ [6, 14], much larger than $n^{-\beta/(\beta-1)}$, so at a typical time no spacing is below ε and the cover time is a dynamical extreme. Between $n^{-\beta/(\beta-1)}$ and n^{-1} lies the intermediate regime of Theorem D; there the infinite-volume dynamics has no spectral gap [22], and the required averaging must come from decorrelation in time rather than from a gap.

1.5. Method and related work. Cover times of random walks have Gumbel fluctuations by Poisson approximation of the number of unvisited sites [1, 5, 11]. For interacting particles the difficulty is joint survival of several spacings. We avoid couplings with independent Bessel processes and work with the Dirichlet form of the n -particle dynamics.

- (a) Survival of the spacings in S up to time t is avoidance of $A_S = \bigcup_{i \in S} \{g_i \leq \varepsilon\}$. Its probability lies between $\pi(f^2)e^{-t\mathcal{E}(f)/\pi(f^2)}$ for test functions f vanishing on A_S , by Jensen's inequality for the spectral measure, and $e^{-\lambda_{A_S} t}$, with λ_{A_S} the principal Dirichlet eigenvalue (Lemma 3.1). Matching the two to relative precision $o(1/\log n)$ gives the Gumbel law through factorial moments (Theorem 3.4). With a spectral gap, the killed semigroup has a single eigenvalue below the gap, as in Aldous–Brown [2], and multi-time survival factorizes; this gives the point process (Theorem 3.5). The gap of Dyson Brownian motion follows from Brascamp–Lieb (Proposition 2.1).
- (b) The gradients of non-adjacent spacings are orthogonal, so the energy of a function vanishing on A_S dominates the sum of its energies along these directions. Adjacent spacings meet at angle $2\pi/3$; the Gram matrix of a run of adjacent directions has norm below 2, which removes clusters (Proposition 4.2).
- (c) Along the direction of one spacing the Gibbs density is $g^\beta \Phi(g, y)$ with Φ decreasing in g (Lemma 2.2). This gives the exact upper bound $\mathbb{P}(g_i \leq x) \leq a_i x^{\beta+1}/(\beta+1)$ and reduces the lower bound to a moment of the inverse square distances seen from a collision, computed at $\beta = 2$ from the Palm kernel of a determinantal process.

Hitting times of rare sets in reversible chains are approximately exponential [2, 3]; capacities and principal Dirichlet eigenvalues of small sets are the tools of metastability [8]. Small spacings of random matrices at a fixed time are described in [6, 13, 14, 7]. Poisson point processes for extremes and factorial moments are classical [20, 18].

1.6. Organization. Section 2 sets up the Dyson processes and collision coefficients. Section 3 proves the abstract limit theorems, Section 4 the capacity estimates, and Section 5 Theorems A–C. Section 6 treats the circular ensemble, Section 7 variable density, Section 8 the intermediate regime and Theorem D, and Section 9 clusters and open problems.

2. DYSON PROCESSES AS REVERSIBLE DIFFUSIONS

2.1. The two models. *Hermite.* Let $W = \{\lambda \in \mathbb{R}^n : \lambda_1 < \dots < \lambda_n\}$ and

$$\mu(d\lambda) = Z^{-1} e^{-V(\lambda)} d\lambda, \quad V(\lambda) = \frac{\beta n}{4} \sum_k \lambda_k^2 - \beta \sum_{j < k} \log(\lambda_k - \lambda_j).$$

The generator of (1) is $Lf = \frac{1}{\beta n}(\Delta f - \nabla V \cdot \nabla f)$, with Dirichlet form

$$(7) \quad \mathcal{E}(f) = \frac{1}{\beta n} \int |\nabla f|^2 d\mu.$$

For $\beta \geq 1$, (1) has a unique strong solution which never leaves W [4, Prop. 4.3.5]; it is the μ -symmetric diffusion associated with the closure $(\mathcal{E}, \mathcal{F})$ of (7) on $C_c^\infty(W)$, and since ∂W is polar, $\mathcal{F}$ contains the bounded Lipschitz functions on W [17].

Circular. Let $\theta_1 < \dots < \theta_n < \theta_1 + 2\pi$, $g_k = \theta_{k+1} - \theta_k$ for $k < n$ and $g_n = \theta_1 + 2\pi - \theta_n$. The spacings depend only on the projection of θ to $H = \{\sum_k \theta_k = 0\}$, and the projected process is a diffusion on the convex set $W^\circ = \{\theta \in H : \theta_1 < \dots < \theta_n < \theta_1 + 2\pi\}$, reversible for

$$\mu(d\theta) = Z^{-1} e^{-V(\theta)} d\theta, \quad V(\theta) = -\beta \sum_{j < k} \log \left| 2 \sin \frac{\theta_k - \theta_j}{2} \right|,$$

with Lebesgue measure on H and Dirichlet form (7), gradients taken in H [10]. The spacings have the law of the circular β -ensemble, invariant under cyclic relabelling.

2.2. Spectral gap.

Proposition 2.1. *For $\beta \geq 1$, $n \geq 2$ and $f \in \mathcal{F}$, $\text{Var}_\mu(f) \leq c_P^{-1} \mathcal{E}(f)$, with $c_P = \frac{1}{2}$ in the Hermite case and $c_P = \frac{1}{4}$ in the circular case.*

Proof. In both cases μ has density e^{-V} on an open convex set, with V smooth there and tending to $+\infty$ at the boundary. By the Brascamp–Lieb inequality [9], $\nabla^2 V \geq \rho$ implies $\text{Var}_\mu(f) \leq \rho^{-1} \int |\nabla f|^2 d\mu = \rho^{-1} \beta n \mathcal{E}(f)$. In the Hermite case $\nabla^2 V \geq \frac{\beta n}{2}$, since $-\log(\lambda_k - \lambda_j)$ is convex. In the circular case, for $v \in H$,

$$v^\top \nabla^2 V v = \frac{\beta}{4} \sum_{j < k} \frac{(v_k - v_j)^2}{\sin^2 \frac{\theta_k - \theta_j}{2}} \geq \frac{\beta}{4} \sum_{j < k} (v_k - v_j)^2 = \frac{\beta n}{4} |v|^2. \quad \square$$

2.3. Spacing coordinates and collision coefficients. Fix a spacing i . Let $w_i = \frac{1}{2}(e_{i+1} - e_i)$ (with $e_{n+1} = e_1$ in the circular case), so $\partial_{w_i} g_i = 1$ and $|w_i|^2 = \frac{1}{2}$. Write a configuration as (g, y) with $g = g_i$: in the Hermite case $y = (\lambda_1, \dots, \lambda_{i-1}, m, \lambda_{i+2}, \dots, \lambda_n)$, $m = \frac{1}{2}(\lambda_i + \lambda_{i+1})$, so $\lambda_i = m - \frac{g}{2}$, $\lambda_{i+1} = m + \frac{g}{2}$, with Jacobian one; in the circular case the same in an affine chart of H , the constant Jacobian absorbed in Z . Then $\partial_g = \partial_{w_i}$ and

$$(8) \quad e^{-V} = g^\beta \Phi_i(g, y),$$

where, with j ranging over the indices other than $i, i+1$ and up to factors free of g ,

$$\begin{aligned} \Phi_i(g, y) &= e^{-\frac{\beta n}{8} g^2} \prod_j \left| (m - \lambda_j)^2 - \frac{g^2}{4} \right|^\beta, \\ \Phi_i(g, y) &= \left(\frac{\sin(g/2)}{g/2} \right)^\beta \prod_j \left| 4 \sin^2 \frac{m - \theta_j}{2} - 4 \sin^2 \frac{g}{4} \right|^\beta \end{aligned}$$

in the Hermite and circular cases. The admissible g are $0 < g < g_{\max}(y)$, where $g_{\max}(y)/2$ is the distance from m to the nearest other particle. Let

$$D_i(y) = \sum_j (m - \lambda_j)^{-2} \quad \text{or} \quad D_i(y) = \sum_j \left(2 \sin \frac{m - \theta_j}{2} \right)^{-2}.$$

Lemma 2.2. *For admissible (g, y) , $\Phi_i(g, y) \leq \Phi_i(0, y)$. If moreover $g^2 D_i(y) \leq 2$, then $\Phi_i(g, y) \geq \Phi_i(0, y) \exp(-\frac{\beta g^2}{2}(D_i(y) + n))$.*

Proof. In the Hermite case $\Phi_i(g, y)/\Phi_i(0, y) = e^{-\beta n g^2/8} \prod_j (1 - u_j)^\beta$ with $u_j = g^2/(4(m - \lambda_j)^2) \in [0, 1)$, which is at most one; if $g^2 D_i \leq 2$ then $u_j \leq \frac{1}{2}$ and $\log(1 - u) \geq -2u$ gives the lower bound. In the circular case the ratio is $(\frac{\sin(g/2)}{g/2})^\beta \prod_j (1 - u_j)^\beta$ with $u_j = \sin^2 \frac{g}{4} / \sin^2 \frac{m - \theta_j}{2} \in [0, 1)$; now $2 \sum_j u_j \leq \frac{g^2}{2} D_i$, and $\frac{\sin x}{x} \geq e^{-x^2/3}$ for $0 \leq x \leq 1$. $\square$

Lemma 2.3. *Let $c_i(y) = Z^{-1} \Phi_i(0, y)$ for y with m strictly between its neighbours, and $a_i = \int c_i dy$. Then $0 < a_i < \infty$, $p_i(x) \leq a_i x^\beta$ for all $x > 0$, and $x^{-\beta} p_i(x) \rightarrow a_i$ as $x \downarrow 0$.*

Proof. Up to a constant, c_i is the density of an $(n - 1)$ -particle Gibbs measure with a charge 2β at m , hence integrable. By (8), $p_i(x) = x^\beta \int Z^{-1} \Phi_i(x, y) \mathbf{1}\{x < g_{\max}(y)\} dy$; the integrand is at most $c_i(y)$ and tends to it as $x \downarrow 0$. $\square$

The collision law $\hat{\nu}_i(dy) = a_i^{-1} c_i(y) dy$ is the law of the other particles given that λ_i and λ_{i+1} have merged.

Hypothesis 2.4. There are $C_0 \geq 1$ and $\kappa \geq 0$ such that for all large n and all $i \in I_n$:

- (P1) $\max_{I_n} a_i \leq C_0 \min_{I_n} a_i$;
- (P2) $C_0^{-1} n^{\beta+1-\kappa} \leq a_i \leq C_0 n^{\beta+1+\kappa}$;
- (P3) $\int D_i d\hat{\nu}_i \leq C_0 n^{2+\kappa}$.

For the circular ensemble (P1) is an equality and (P2) holds with $\kappa = 0$ by (5); (P3) holds with $\kappa = 0$ for the circular unitary ensemble (Proposition 6.3).

3. RARE HITTING TIMES IN REVERSIBLE DIFFUSIONS

Let X be a diffusion on E , symmetric with respect to a probability measure π , with regular Dirichlet form $(\mathcal{E}, \mathcal{F})$, started from π . For a closed set A let $T_A = \inf\{t \geq 0 : X_t \in A\}$, $D = E \setminus A$, $\mathcal{F}_A = \{f \in \mathcal{F} : \tilde{f} = 0 \text{ q.e. on } A\}$, and

$$\lambda_A = \inf\{\mathcal{E}(f)/\pi(f^2) : f \in \mathcal{F}_A, f \neq 0\}.$$

The part of X on D is associated with $(\mathcal{E}, \mathcal{F}_A)$, and $\mathbb{P}_\pi(T_A > t) = \langle \mathbf{1}_D, P_t^D \mathbf{1}_D \rangle_\pi$ with P^D the killed semigroup [17, Thms. 4.4.2–4.4.3].

Lemma 3.1. *For every closed A , every $t \geq 0$ and every $f \in \mathcal{F}_A$ with $0 \leq f \leq 1$,*

$$\pi(f^2) \exp\left(-t \frac{\mathcal{E}(f)}{\pi(f^2)}\right) \leq \mathbb{P}_\pi(T_A > t) \leq e^{-\lambda_A t}.$$

Moreover $\mathbb{E}_\pi T_A \leq 1/\lambda_A$ and $\mathbb{E}_\pi T_A \geq \pi(f)^2/\mathcal{E}(f)$.

Proof. Let ν be the spectral measure of $\mathbf{1}_D$ for the killed generator; it is carried by $[\lambda_A, \infty)$ and has mass $\pi(D) \leq 1$, so $\mathbb{P}_\pi(T_A > t) = \int e^{-st} \nu(ds) \leq e^{-\lambda_A t}$ and $\mathbb{E}_\pi T_A \leq 1/\lambda_A$. Since P_t^D preserves positivity and $0 \leq f \leq \mathbf{1}_D$, $\langle \mathbf{1}_D, P_t^D \mathbf{1}_D \rangle \geq \langle f, P_t^D f \rangle = \int e^{-st} \nu_f(ds)$, where ν_f has mass $\pi(f^2)$ and first moment $\mathcal{E}(f)$; Jensen's inequality gives the lower bound. Finally $u = \int_0^\infty P_t^D \mathbf{1}_D dt$ satisfies $\mathcal{E}(u) = \langle u, \mathbf{1}_D \rangle = \mathbb{E}_\pi T_A$ and $\pi(f) = \mathcal{E}(f, u) \leq \mathcal{E}(f)^{1/2} \mathcal{E}(u)^{1/2}$. $\square$

Lemma 3.2. *Assume $\text{Var}_\pi(f) \leq c_P^{-1}\mathcal{E}(f)$ on $\mathcal{F}$, and let $0 < \lambda_A < c_P$. Then λ_A is a simple isolated eigenvalue of the killed generator with eigenfunction $\varphi_A \geq 0$, $\|\varphi_A\| = 1$, the rest of the spectrum lies in $[c_P, \infty)$, and with $c_A = \langle \mathbf{1}_D, \varphi_A \rangle^2$,*

$$0 \leq \mathbb{P}_\pi(T_A > t) - c_A e^{-\lambda_A t} \leq (1 - c_A)e^{-c_P t}, \quad c_A \geq \lambda_A (\mathbb{E}_\pi T_A - c_P^{-1}).$$

Proof. Every two-dimensional subspace of $\mathcal{F}_A$ contains $f \neq 0$ with $\pi(f) = 0$, for which $\mathcal{E}(f) \geq c_P \pi(f^2)$; by min-max the spectral projection on $[0, c_P)$ has rank at most one. If φ is an eigenfunction, $|\varphi| \in \mathcal{F}_A$ has the same Rayleigh quotient, so $\varphi = \pm|\varphi|$. With ν as above, $\nu(\{\lambda_A\}) = c_A$ and $\nu([c_P, \infty)) \leq 1 - c_A$; this gives the first bound, and integrating in t gives $\mathbb{E}_\pi T_A \leq c_A/\lambda_A + c_P^{-1}$. $\square$

Lemma 3.3. *In the setting of Lemma 3.2, let $A_1 \supseteq \dots \supseteq A_k$ be closed with $0 < \lambda_{A_l} < c_P$, $0 = t_0 < t_1 < \dots < t_k$, $\Delta_l = t_l - t_{l-1}$. Then*

$$\mathbb{P}_\pi(X_s \notin A_l \text{ for } s \leq t_l, l \leq k) = \Theta \prod_l e^{-\lambda_{A_l} \Delta_l} + \rho,$$

with $1 - 5 \sum_l (1 - c_{A_l}) \leq \Theta \leq 1$ and $|\rho| \leq 2^k e^{-c_P \min_l \Delta_l}$.

Proof. Let $Q_l f = P_{\Delta_l}^{D_l}(f \mathbf{1}_{D_l})$. By the Markov property the probability is $\langle 1, Q_1 \dots Q_k 1 \rangle$. By Lemma 3.2, $Q_l = e^{-\lambda_{A_l} \Delta_l} \varphi_l \langle \varphi_l, \cdot \rangle + R_l$ with $\|R_l\| \leq e^{-c_P \Delta_l}$. The term without R_l 's is $\prod_l e^{-\lambda_{A_l} \Delta_l} \Theta$, $\Theta = \langle 1, \varphi_1 \rangle \langle \varphi_1, \varphi_2 \rangle \dots \langle \varphi_k, 1 \rangle$, and each of the other $2^k - 1$ terms is at most $e^{-c_P \min_l \Delta_l}$. All inner products lie in $[0, 1]$; $\|\varphi_l - 1\|^2 \leq 2(1 - c_{A_l})$ gives $\langle \varphi_l, \varphi_{l+1} \rangle \geq 1 - 2(1 - c_{A_l}) - 2(1 - c_{A_{l+1}})$, and $\langle 1, \varphi_l \rangle \geq c_{A_l}$. $\square$

Consider now a sequence of such diffusions with finite index sets I_n , $N_n = |I_n| \rightarrow \infty$, a graph on I_n of degree at most d , closed sets A_i and rates $\Lambda_i > 0$. For $S \subset I_n$ let $A_S = \bigcup_{i \in S} A_i$, $\lambda_S = \lambda_{A_S}$, $\Lambda_S = \sum_{i \in S} \Lambda_i$; S is *separated* if it contains no two neighbours. Let $\tau_i = T_{A_i}$, $\Lambda_{\max} = \max \Lambda_i$, $\Lambda_{\min} = \min \Lambda_i$, and t_n, x_n as in (6). Consider the conditions:

- (A) $\Lambda_{\max} \leq C_0 \Lambda_{\min}$;
- (B) for every m there are $\eta_n = \eta_n(m)$ with $\eta_n \log N_n \rightarrow 0$ such that for all S with $|S| \leq m$,

$$(9) \quad \lambda_S \geq (1 - \eta_n) \sum_C \left(1 + \frac{1}{3} \mathbf{1}_{\{|C| \geq 2\}}\right) \min_{i \in C} \Lambda_i,$$

the sum over the connected components C of S ;

- (C) for every separated S with $|S| \leq m$ there is $f_S \in \mathcal{F}_{A_S}$, $0 \leq f_S \leq 1$, with $\pi(f_S^2) \geq 1 - \eta_n$ and $\mathcal{E}(f_S) \leq (1 + \eta_n) \Lambda_S \pi(f_S^2)$.

Theorem 3.4. *Under (A)–(C), for every $x \in \mathbb{R}$ the number of i with $\tau_i > t_n(x)$ converges in law to a Poisson variable with mean e^{-x} . Hence $\mathbb{P}(\max_i \tau_i \leq t_n(x)) \rightarrow \exp(-e^{-x})$, the k -th largest $\tau_{(k)}$ satisfies $\mathbb{P}(\tau_{(k)} \leq t_n(x)) \rightarrow e^{-e^{-x}} \sum_{j < k} e^{-jx}/j!$, and $\sup_t |\mathbb{P}(\tau_i > t) - e^{-\Lambda_i t}| \leq 3\eta_n(1)$ for all i .*

Proof. By Lemma 3.1, (B) and (C), for separated S with $|S| \leq m$,

$$(10) \quad (1 - \eta_n) e^{-(1+\eta_n)\Lambda_S t} \leq \mathbb{P}(T_{A_S} > t) \leq e^{-(1-\eta_n)\Lambda_S t}.$$

For $S = \{i\}$ this gives the last claim. Since $N_n e^{-\Lambda_{\max} t} \leq \sum_i e^{-\Lambda_i t} \leq N_n e^{-\Lambda_{\min} t}$,

$$(11) \quad (\log N_n + x)/\Lambda_{\max} \leq t_n(x) \leq (\log N_n + x)/\Lambda_{\min}.$$

Fix x , $t = t_n(x)$, $W = \#\{i : \tau_i > t\}$ and $m \geq 1$. Then $\mathbb{E}(W)_m = \sum \mathbb{P}(T_{A_S} > t)$ over ordered m -tuples of distinct indices, S their set. For separated S , (10) and $\Lambda_S t \leq m C_0 (\log N_n + x)$ give $\mathbb{P}(T_{A_S} > t) = (1 + o(1)) \prod_{i \in S} e^{-\Lambda_i t}$ uniformly. Summed over all

tuples, with repetitions allowed, the product equals $(\sum_i e^{-\Lambda_i t})^m = e^{-mx}$; tuples with a repeated index or two neighbours contribute to this sum at most $m^2(d+1)e^{-\Lambda_{\min} t}e^{-(m-1)x} \rightarrow 0$, since $\Lambda_{\min} t \geq (\log N_n + x)/C_0$.

For non-separated S , $\mathbb{P}(T_{AS} > t) \leq e^{-\lambda_S t} \leq \prod_C e^{-(1-\eta_n)w_C t}$, where $w_C = (1 + \frac{1}{3}\mathbf{1}_{|C| \geq 2}) \min_C \Lambda_i$. Group the tuples by the shape of their decomposition into components; the sum over placements factorizes over components. A singleton contributes $\sum_i e^{-(1-\eta_n)\Lambda_i t} \leq e^{\eta_n \Lambda_{\max} t} e^{-x} = O(1)$. A component of size $s \geq 2$ is a connected set containing its minimizing index, and there are at most $(ed)^s$ such sets per vertex, so it contributes at most $(ed)^s e^{-x} e^{\eta_n \Lambda_{\max} t} \max_i e^{-(\frac{1}{3}-\frac{4}{3}\eta_n)\Lambda_i t} \rightarrow 0$. Hence $\mathbb{E}(W)_m \rightarrow e^{-mx}$ for every m , and W converges to $\text{Poisson}(e^{-x})$. The statements on order statistics follow from $\{\tau_{(k)} \leq t\} = \{W < k\}$. $\square$

Theorem 3.5. *Assume (A)–(C), $\text{Var}_\pi(f) \leq c_P^{-1}\mathcal{E}(f)$ with $c_P \geq c > 0$ for all n , and $\Lambda_{\max} \log N_n \rightarrow 0$. Then for every $x_0 \in \mathbb{R}$, $\sum_i \delta_{x_n(\tau_i)}$ converges in distribution on $(x_0, \infty]$ to the Poisson process with intensity $e^{-x}dx$.*

Proof. For separated S , $|S| \leq m$: by (C) and Lemma 3.1, $\lambda_S \leq (1 + \eta_n)\Lambda_S$ and, since $\pi(f_S) \geq \pi(f_S^2)$, $\mathbb{E}_\pi T_{AS} \geq (1 - \eta_n)/((1 + \eta_n)\Lambda_S)$; with (B), $\lambda_S < c$ for large n and Lemma 3.2 gives $c_{AS} \geq 1 - \eta'_n$, $\eta'_n \rightarrow 0$.

Fix $x_0 < a_1 < b_1 \leq a_2 < \dots < b_k \leq \infty$, $J_l = (t_n(a_l), t_n(b_l)]$, $M_l = \#\{i : \tau_i \in J_l\}$. We show $\mathbb{E} \prod_l (M_l)_{m_l} \rightarrow \prod_l (e^{-a_l} - e^{-b_l})^{m_l}$; by the method of moments the M_l then converge jointly to independent Poisson variables, which characterizes convergence to the Poisson process [18, Ch. 4]. The expectation is a sum over families of distinct indices of $\mathbb{P}(\tau_i \in J_{l(i)}, i \in S)$. Since $t'_n \geq 1/\Lambda_{\max}$, distinct endpoints satisfy $|t_n(a) - t_n(b)| \geq |a - b|/\Lambda_{\max}$, and $t_n(a_1) \geq (\log N_n + a_1)/\Lambda_{\max}$.

For separated S , expand $\mathbf{1}\{\tau \in J_l\} = \mathbf{1}\{\tau > t_n(a_l)\} - \mathbf{1}\{\tau > t_n(b_l)\}$. Each resulting term $\mathbb{P}(\tau_i > u_i, i \in S)$ is the probability of avoiding a decreasing sequence of sets A_{S_p} on $[0, s_p]$, $s_1 < s_2 < \dots$ the distinct values of the u_i . By Lemma 3.3 it equals $(1 + o(1)) \exp(-\sum_i \Lambda_i u_i) + O(2^m e^{-ch/\Lambda_{\max}})$ for some $h > 0$, because $\lambda_{S_p} = (1 + O(\eta_n))\Lambda_{S_p}$ and $\sum_i \Lambda_i u_i = O(\log N_n)$. Summing back, the main term is $\prod_i (e^{-\Lambda_i t_n(a_{l(i)})} - e^{-\Lambda_i t_n(b_{l(i)})})$, whose sum over all families with repetitions is $\prod_l (e^{-a_l} - e^{-b_l})^{m_l}$ by (6). Non-distinct or non-separated families, the $o(1)$ terms, and the terms $N_n^m 2^m e^{-ch/\Lambda_{\max}}$ (which vanish because $\Lambda_{\max} \log N_n \rightarrow 0$) are negligible exactly as in the proof of Theorem 3.4, with $t = t_n(a_1)$. $\square$

4. CAPACITY OF SPACING EVENTS

Return to the Dyson processes, with $A_i = \{g_i \leq \varepsilon\}$, i and $i \pm 1$ neighbours (indices mod n in the circular case), and hazards (3). Constants C depend only on β , C_0 and m .

Proposition 4.1. *Let S be separated, $|S| \leq m$, $0 < \varepsilon < R$, and $f = \prod_{i \in S} (1 - h(g_i))$ with $h = 1$ on $[0, \varepsilon]$, $h(g) = \frac{g^{1-\beta} - R^{1-\beta}}{\varepsilon^{1-\beta} - R^{1-\beta}}$ on $[\varepsilon, R]$, $h = 0$ on $[R, \infty)$. Then $f \in \mathcal{F}_{AS}$, $0 \leq f \leq 1$,*

$$\mathcal{E}(f) \leq \frac{\Lambda_S}{1 - (\varepsilon/R)^{\beta-1}}, \quad \pi(f^2) \geq 1 - \sum_{i \in S} \frac{a_i R^{\beta+1}}{\beta + 1}.$$

Proof. f is bounded, Lipschitz and vanishes on A_S . Since $\nabla g_i = 2w_i$ and $w_i \perp w_j$ for non-neighbours, $|\nabla f|^2 \leq 2 \sum_{i \in S} h'(g_i)^2$. By Lemma 2.3, $\int h'(g_i)^2 d\mu \leq a_i \int_\varepsilon^R h'(g)^2 g^\beta dg = a_i(\beta - 1)\varepsilon^{\beta-1}/(1 - (\varepsilon/R)^{\beta-1})$. Finally $f = 1$ unless some $g_i < R$, and $\mu(g_i < R) \leq a_i R^{\beta+1}/(\beta + 1)$. $\square$

For a run C of s consecutive indices let $\mu_C = 1 + \cos \frac{\pi}{s+1}$, and let $\mathcal{C}(S)$ be the decomposition of S into runs.

Proposition 4.2. *Assume Hypothesis 2.4. Let $K \geq 1$ and $R > 0$ with $8K(nR)^2 \leq 1$. For every $S \subset I_n$ with $|S| \leq m$,*

$$\lambda_S \geq (1 - \eta) \sum_{C \in \mathcal{C}(S)} \mu_C^{-1} \sum_{i \in C} \Lambda_i, \quad \eta = C \left(K(nR)^2 + \frac{n^\kappa}{K} + \left(\frac{\Lambda_{\max} n^\kappa}{(nR)^{\beta+1}} \right)^{1/2} + \Lambda_{\max} \right).$$

Proof. Geometry. Let $u_i = \sqrt{2}w_i$. For $|i - j| \geq 2$, $u_i \perp u_j$, and $\langle u_i, u_{i+1} \rangle = -\frac{1}{2}$; the Gram matrix of a run C is tridiagonal with largest eigenvalue μ_C , and different runs span orthogonal subspaces. Hence $|v|^2 \geq \sum_C \mu_C^{-1} \sum_{i \in C} \langle u_i, v \rangle^2$, and with $\langle u_i, \nabla f \rangle^2 = 2(\partial_{g_i} f)^2$,

$$(12) \quad \mathcal{E}(f) \geq \sum_C \mu_C^{-1} \sum_{i \in C} \mathcal{E}_i(f), \quad \mathcal{E}_i(f) = \frac{2}{\beta n} \int (\partial_{g_i} f)^2 d\mu.$$

One direction. Let $f \in \mathcal{F}_{A_S}$. If $\pi(f) = 0$, then $\mathcal{E}(f) \geq c_P \pi(f)^2$ and there is nothing to prove; if $\mathcal{E}(f) > 2\Lambda_S \pi(f)^2$, go to the conclusion. Otherwise fix $i \in S$ and $G = \{y : D_i(y) \leq Kn^2\}$. For $y \in G$ and $g \leq 2R$, $g^2 D_i \leq 2$, so (g, y) is admissible and, by Lemma 2.2, $Z^{-1} g^\beta \Phi_i(g, y) \geq e^{-\eta_1} c_i(y) g^\beta$ with $\eta_1 \leq 8\beta K(nR)^2$. For almost every y , $g \mapsto f(g, y)$ is absolutely continuous and vanishes at ε , so for $r \in [R, 2R]$

$$f(r, y)^2 \leq \int_\varepsilon^r (\partial_g f)^2 g^\beta dg \int_\varepsilon^r g^{-\beta} dg \leq \frac{\varepsilon^{1-\beta}}{\beta-1} \int_\varepsilon^r (\partial_g f)^2 g^\beta dg.$$

Averaging over r with weight r^β/M_R , $M_R = \int_R^{2R} r^\beta dr \geq R^{\beta+1}$, and setting $\sigma(dr dy) = M_R^{-1} r^\beta c_i(y) \mathbf{1}_G(y) \mathbf{1}_{[R, 2R]}(r) dr dy$,

$$\mathcal{E}_i(f) \geq e^{-\eta_1} \frac{2(\beta-1)}{\beta n} \varepsilon^{\beta-1} \int f^2 d\sigma, \quad \sigma(1) \geq a_i(1 - \eta_P), \quad \eta_P = \hat{\nu}_i(D_i > Kn^2) \leq \frac{C_0 n^\kappa}{K}.$$

On the slab $\sigma \leq e^{\eta_1} M_R^{-1} \mu$, so by Proposition 2.1, $\|f - \pi(f)\|_{L^2(\sigma)}^2 \leq e^{\eta_1} \mathcal{E}(f)/(c_P M_R) \leq 2e^{\eta_1} \Lambda_S \pi(f)^2/(c_P M_R)$, and

$$\left(\int f^2 d\sigma \right)^{1/2} \geq |\pi(f)| \sqrt{a_i} (\sqrt{1 - \eta_P} - \sqrt{\eta_2}), \quad \eta_2 = \frac{2e^{\eta_1} \Lambda_S}{c_P a_i M_R} \leq \frac{C \Lambda_{\max} n^\kappa}{(nR)^{\beta+1}},$$

using $\Lambda_S/a_i \leq mC_0 \frac{2(\beta-1)}{\beta n} \varepsilon^{\beta-1} \leq mC_0^2 \Lambda_{\max} n^{\kappa-\beta-1}$ by (P1), (P2). Thus $\mathcal{E}_i(f) \geq (1 - \eta_1 - \eta_P - 2\sqrt{\eta_2}) \Lambda_i \pi(f)^2$.

Conclusion. By (12), $\mathcal{E}(f) \geq (1 - \eta') \Lambda_S^{\text{eff}} \pi(f)^2$ with $\Lambda_S^{\text{eff}} = \sum_C \mu_C^{-1} \sum_{i \in C} \Lambda_i \leq \Lambda_S$, in all cases. Since $\pi(f^2) \leq \pi(f)^2 + c_P^{-1} \mathcal{E}(f)$, $\mathcal{E}(f)/\pi(f^2) \geq (1 - \eta' - 4m\Lambda_{\max}) \Lambda_S^{\text{eff}}$. $\square$

Corollary 4.3. *Assume Hypothesis 2.4. Conditions (A)–(C) of Section 3 hold for the sets A_i , $i \in I_n$, with $\eta_n \log n \rightarrow 0$, in each of the cases:*

- (i) $\kappa = 0$ and $\Lambda_{\max}(\log n)^{2\beta+4} \rightarrow 0$;
- (ii) $\Lambda_{\max} \leq n^{-\delta}$ and $\kappa \leq \delta/(4(\beta+1))$.

Proof. (A) is (P1). For runs of length $s \geq 2$, $s/\mu_C \geq \frac{4}{3}$, so Proposition 4.2 gives (B). In case (i) take $L = \log n$, $K = L^2$, $nR = L^{-2}$: then $\eta \leq C(L^{-2} + (\Lambda_{\max} L^{2\beta+2})^{1/2} + \Lambda_{\max})$ and $\eta L \rightarrow 0$. In case (ii) take $K = n^\theta$, $nR = n^{-\theta}$, $\theta = \delta/(2(\beta+1))$: then $\eta \leq Cn^{-\theta/2}$. For (C) use Proposition 4.1 with the same R : $\sum_i a_i R^{\beta+1} \leq mC_0 n^\kappa (nR)^{\beta+1}$, and since $(n\varepsilon)^{\beta-1} \leq C\Lambda_{\max} n^{\kappa-1}$ by (P2), $(\varepsilon/R)^{\beta-1} = (n\varepsilon)^{\beta-1} (nR)^{1-\beta}$ is $O(n^{-1/2})$ in both cases. $\square$

5. PROOFS OF THEOREMS A–C

Proof of Theorems B and C. By Corollary 4.3, (A)–(C) hold with $\eta_n \log n \rightarrow 0$; by Proposition 2.1 the spectral gap is at least $\frac{1}{4}$; and $\Lambda_{\max} \log n \rightarrow 0$. Theorems 3.4 and 3.5 give Theorem C. In the circular case all hazards equal λ_n , so $x_n(\tau) = \lambda_n \tau - \log n$, which is Theorem B. $\square$

Proof of Theorem A. By Proposition 6.1, $a_2(n) = n(n^2 - 1)/(24\pi)$, so (P1)–(P2) hold with $\kappa = 0$, and (P3) holds with $\kappa = 0$ by Proposition 6.3. The hazard is $\lambda_n = (n^2 - 1)\varepsilon_n/(24\pi)$, and $\lambda_n(\log n)^8 \rightarrow 0$; apply Theorem B. $\square$

6. THE CIRCULAR ENSEMBLE

Proposition 6.1. *The collision coefficient of the circular β -ensemble is (5).*

Proof. Let $P = Z_n^{-1} \prod_{j < k} |e^{i\theta_j} - e^{i\theta_k}|^\beta$ on $[0, 2\pi)^n$, where $Z_n = (2\pi)^n \Gamma(1 + \frac{n\beta}{2}) / \Gamma(1 + \frac{\beta}{2})^n$ [16]. All factors of P are bounded, so by dominated convergence and rotation invariance

$$\lim_{x \downarrow 0} x^{-\beta} \rho_2(\phi, \phi + x) = \frac{n(n-1)(2\pi)^{n-2}}{Z_n} M_{n-2}(\beta, \beta, \frac{\beta}{2}),$$

where, by Morris' identity [16],

$$\begin{aligned} M_N(a, b, \lambda) &= \int_{[-\pi, \pi]^N} \prod_j e^{\frac{i}{2}(a-b)\theta_j} |1 + e^{i\theta_j}|^{a+b} \prod_{j < k} |e^{i\theta_j} - e^{i\theta_k}|^{2\lambda} \frac{d\theta}{(2\pi)^N} \\ &= \prod_{j=0}^{N-1} \frac{\Gamma(\lambda j + a + b + 1) \Gamma(\lambda(j+1) + 1)}{\Gamma(\lambda j + a + 1) \Gamma(\lambda j + b + 1) \Gamma(1 + \lambda)}. \end{aligned}$$

For small x , pairs at distance x are consecutive up to $O(x^{2\beta+1})$, and the n spacings are identically distributed, so $np(x) = 2\pi\rho_2(0, x) + O(x^{2\beta+1})$, which gives (5). At $\beta = 2$ the product telescopes, $\prod_{j=0}^{n-3} \frac{(j+4)(j+3)}{j+2} = \frac{n(n+1)!}{12}$, and $a_2(n) = n(n^2 - 1)/(24\pi)$. $\square$

Remark 6.2. By Stirling's formula, $(2\pi)^{\beta+1} n^{-\beta-1} a_\beta(n) \rightarrow C_\beta$, where

$$C_\beta = (\pi\beta)^\beta \frac{\Gamma(1 + \frac{\beta}{2})^3}{\Gamma(1 + \beta) \Gamma(1 + \frac{3\beta}{2})}$$

is the small-distance coefficient of the two-point function of Sine_β [15, Ch. 13]: $C_1 = \pi^2/6$, $C_2 = \pi^2/3$, $C_4 = 16\pi^4/135$, and $\kappa_\beta = C_\beta/(\beta + 1)$. Hence $\lambda_n \sim \frac{2(\beta-1)}{\beta} C_\beta (2\pi)^{-\beta-1} n^\beta \varepsilon^{\beta-1}$.

As a law of $n - 2$ unordered points, the collision law of the circular ensemble is the law Q_m with density proportional to $\prod_j |e^{im} - e^{i\theta_j}|^{2\beta} \prod_{j < k} |e^{i\theta_j} - e^{i\theta_k}|^\beta$, averaged over a uniform m .

Proposition 6.3. *For the circular unitary ensemble, $\int D_i d\hat{v}_i \leq n^2$.*

Proof. The CUE is determinantal with kernel $K(x, y) = \langle v_y, v_x \rangle$, $v_x = (2\pi)^{-1/2} (e^{-ikx})_{k < n} \in \mathbb{C}^n$, and $\rho_3(p, q, z)/\rho_2(p, q) = \text{dist}(v_z, \text{span}\{v_p, v_q\})^2$. Letting $p, q \rightarrow m$ as in Proposition 6.1, the one-point density of Q_m is $\rho^{(m)}(z) = \text{dist}(v_z, \text{span}\{v_m, \partial v_m\})^2$. Hence $\rho^{(m)}(z) \leq |v_z|^2 = \frac{n}{2\pi}$ and, for $u = z - m \in [-\pi, \pi]$, $\rho^{(m)}(z) \leq |v_z - v_m - u\partial v_m|^2 \leq \frac{u^4}{4} \sup |\partial^2 v|^2 \leq \frac{n^5 u^4}{40\pi}$. With $(2 \sin \frac{u}{2})^{-2} \leq \pi^2/(4u^2)$,

$$\int D dQ_m \leq \int_{|u| \leq 1/n} \frac{n^5 u^4}{40\pi} \cdot \frac{\pi^2}{4u^2} du + \int_{1/n \leq |u| \leq \pi} \frac{n}{2\pi} \cdot \frac{\pi^2}{4u^2} du \leq \frac{\pi n^2}{240} + \frac{\pi n^2}{4} \leq n^2. \quad \square$$

Remark 6.4. For the Gaussian unitary ensemble the same argument with the Hermite kernel bounds the inverse square distances seen from a collision at a bulk point by Cn^2 . Hypothesis 2.4 concerns the collision of a labelled spacing, so it also needs a collision of the i -th spacing to occur within $n^{-1+o(1)}$ of γ_i (rigidity of the conditioned ensemble) and $a_i \asymp n^{\beta+1}$; both are expected for every β .

7. VARIABLE DENSITY AND THE EXPLICIT CENTERING

Proposition 7.1. *Let $0 < a < b < 1$, $I_n = \{i : an \leq i \leq bn\}$, and $\Lambda_i = r_n \psi(i/n)(1 + o(1/\log n))$ uniformly, where $\psi \in C^2[a, b]$ is positive and attains its minimum ψ_* only on a nonempty $E_* \subseteq \{a, b\}$, with $\psi' \neq 0$ on E_* . With $c_\psi = \sum_{e \in E_*} |\psi'(e)|^{-1}$,*

$$\psi_* r_n t_n(x) = \log n - \log \log n + \log(\psi_* c_\psi) + x + o(1).$$

Proof. Put $s = r_n t_n(x) \asymp \log n$ by (11). The errors in Λ_i change $\Lambda_i t_n(x)$ by $o(1)$. Since $e^{-s\psi}$ varies on the scale $1/s \gg 1/n$, $\sum_{I_n} e^{-s\psi(i/n)} = (1 + o(1))n \int_a^b e^{-s\psi}$, and Laplace's method gives $\int_a^b e^{-s\psi} = e^{-s\psi_*} c_\psi s^{-1} (1 + O(s^{-1}))$. Thus $s\psi_* = \log n + \log c_\psi - \log s + x + o(1)$ and $\log s = \log \log n - \log \psi_* + o(1)$. $\square$

Corollary 7.2. *Let $\beta = 2$, $I_n = \{i : \frac{n}{4} \leq i \leq \frac{3n}{4}\}$, assume the hypotheses of Theorem C and $a_i = \frac{\pi^2}{3} (n \varrho(\gamma_i))^3 (1 + o(1/\log n))$ uniformly. Then $\mathbb{P}(\Lambda_* \tau_{\text{cov}} - \log n + \log \log n - \log c_* \leq x) \rightarrow \exp(-e^{-x})$, with Λ_* , c_* as in the introduction.*

Proof. Here $\Lambda_i = r_n \psi(i/n)(1 + o(1/\log n))$ with $r_n = \frac{\pi^2}{3} n^2 \varepsilon$ and $\psi(u) = \varrho(\gamma(u))^3$, $\gamma(u)$ the u -quantile of ϱ . Since $\gamma' = 1/\varrho(\gamma)$, $\psi' = 3\varrho\varrho'(\gamma)$; ψ is minimal exactly at $\frac{1}{4}$ and $\frac{3}{4}$, with $|\psi'| = 3\varrho(\gamma_*)|\varrho'(\gamma_*)|$ at both. So $\psi_* c_\psi = 2\varrho(\gamma_*)^2/(3|\varrho'(\gamma_*)|) = c_*$, and Theorem C with Proposition 7.1 concludes. $\square$

The correction is absent for constant hazards, as in the circular ensemble or on mesoscopic windows of $o(n/\log n)$ indices.

8. THE INTERMEDIATE REGIME

Proof of Theorem D. Condition (A) is (P1) and (B) is assumed. For (C), Proposition 4.1 with $nR = (\log n)^{-1}$ gives $\eta_n \leq C((\log n)^{-\beta-1} + (n\varepsilon \log n)^{\beta-1})$, and $\eta_n \log n \rightarrow 0$ because $n\varepsilon(\log n)^{\beta/(\beta-1)} \rightarrow 0$. Apply Theorem 3.4. $\square$

Theorem 3.4 needs no spectral gap: the upper bound in (10) is spectral and the lower bound is Jensen's inequality. Only (9) uses the gap, through Proposition 4.2. The next lemma isolates what remains.

Lemma 8.1. *For a spacing i and y fixed, let $\ell = g_{\max}(y)$, $w(g) = Z^{-1} g^\beta \Phi_i(g, y)$ on $(0, \ell)$, $M(g) = \int_g^\ell w$, and*

$$J_i(y) = \int_\varepsilon^\ell \frac{M(g)}{w(g)} dg, \quad H_i(y) = \frac{2}{\beta n J_i(y)}.$$

Then $\mathcal{E}_i(f) \geq \int H_i f^2 d\mu$ for every $f \in \mathcal{F}_{A_i}$.

Proof. On almost every line, $u(g) = f(g, y)$ is absolutely continuous with $u(\varepsilon) = 0$, so $u(g)^2 \leq \int_\varepsilon^g u'^2 w \int_\varepsilon^g w^{-1}$. Multiplying by $w(g)$ and integrating, Fubini gives $\int_\varepsilon^\ell u^2 w \leq J_i(y) \int_\varepsilon^\ell u'^2 w$; integrate over y , using $f = 0$ for $g \leq \varepsilon$. $\square$

The function H_i is the hazard of the i -th spacing conditionally on the other coordinates. Writing $m(y) = M(0)$ for the mass of the line, for typical y the integral defining $J_i(y)$ is dominated by g near ε , where $w(g) \approx c_i(y)g^\beta$ and $M(g) \approx m(y)$; thus $J_i(y) \approx m(y)\varepsilon^{1-\beta}/((\beta-1)c_i(y))$ and

$$H_i(y) \approx \Lambda_i \frac{c_i(y)}{a_i m(y)}, \quad \int \frac{c_i(y)}{a_i m(y)} m(y) dy = 1 :$$

H_i has mean Λ_i under μ but fluctuates by a factor of order one with the positions of the neighbouring particles. By (12) and Lemma 8.1, the bound (9) follows from the averaging inequality below.

Conjecture 8.2. *Let $\beta > 1$ and $n\varepsilon_n(\log n)^C \rightarrow 0$ for a suitable $C = C(\beta)$. For every m there is η_n with $\eta_n \log n \rightarrow 0$ such that for every $S \subset I_n$ with $|S| \leq m$, every $i \in S$ and every $f \in \mathcal{F}_{A_S}$ with $\mathcal{E}(f) \leq 2\Lambda_S \pi(f^2)$,*

$$\int H_i f^2 d\mu \geq (1 - \eta_n) \Lambda_i \pi(f^2).$$

Proposition 4.2 proves this when $\Lambda_i(\log n)^{2\beta+4} \rightarrow 0$, using the global gap to make f nearly constant. In general a function of energy $O(\Lambda_S)$ varies on the time scale Λ_S^{-1} , while $H_i/\Lambda_i - 1$ is a local observable which, by local equilibration of Dyson Brownian motion [12, 19], decorrelates on the time scale $n^{-1+o(1)}$. A Kipnis–Varadhan bound $|\int (H_i - \Lambda_i) f^2 d\mu| \leq \|H_i - \Lambda_i\|_{H^{-1}} \mathcal{E}(f^2)^{1/2}$ with $\|H_i - \Lambda_i\|_{H^{-1}}^2 \approx \Lambda_i^2 n^{-1+o(1)}$ then suggests a relative error $(\Lambda_i/n)^{1/2} n^{o(1)} \approx (n\varepsilon)^{(\beta-1)/2} n^{o(1)}$. The global gap cannot be replaced by a local one: the infinite-volume Dyson Brownian motion at $\beta = 2$ has no spectral gap [22]. The averaging has to come from decorrelation in time.

9. CLUSTERS AND OPEN PROBLEMS

9.1. Adjacent spacings. Clusters of adjacent spacings are removed in Theorem 3.4 by the factor $\frac{4}{3}$ of (9). Statically they are much rarer: two adjacent spacings are both below s with probability of order $s^{3\beta+2}$, not $s^{2\beta+2}$. At $\beta = 2$ the constant is explicit.

Proposition 9.1. *For the Sine₂ process with unit density, and g_1, g_2 the two spacings following the point at the origin under the Palm measure,*

$$\mathbb{P}(g_1 \leq s, g_2 \leq s) \sim \frac{31\pi^6}{16200} s^8, \quad s \downarrow 0.$$

Proof. Let $K(x, y) = \frac{\sin \pi(x-y)}{\pi(x-y)}$. The Palm measure at 0 is determinantal with kernel $K_0(x, y) = K(x, y) - K(x, 0)K(0, y)$ [21]. Its two-point function $\rho(a, b) = K_0(a, a)K_0(b, b) - K_0(a, b)^2$ vanishes to second order on $\{a = 0\}$, $\{b = 0\}$ and $\{a = b\}$, and expanding the sine kernel at the origin gives $\rho(a, b) = \frac{\pi^6}{135} a^2 b^2 (a-b)^2 + O(|(a, b)|^8)$. By inclusion–exclusion for determinantal processes, $\mathbb{P}(g_1 \leq s, g_2 \leq s)$ equals the integral of ρ over $\{0 < a < s, a < b < a + s\}$ up to terms of higher order, and $\int_0^s \int_0^s a^2 u^2 (a+u)^2 du da = \frac{31}{120} s^8$. $\square$

9.2. Open problems.

- (1) Conjecture 8.2, which by Theorem D gives the Gumbel law throughout $n\varepsilon \ll (\log n)^{-C}$; the point process in that regime also needs multi-time survival bounds without a gap.
- (2) Hypothesis 2.4 for the Hermite ensemble and for $\beta \neq 2$ (Remark 6.4).
- (3) The recurrent case $\beta = 1$: the one-dimensional capacity becomes $\int_\varepsilon^R g^{-1} dg$, the hazard is of order $a_i/(n \log(1/(n\varepsilon)))$, and the cut-off R enters through $\log R$, which defeats the precision $o(1/\log n)$.
- (4) Non-equilibrium initial data, such as Wigner matrices run through the matrix Ornstein–Uhlenbeck flow. Since the cover time exceeds the relaxation time, the same limits are expected; hits occurring before relaxation must be controlled.

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